Probabilistic Machine Learning

Lecture 03: Generative Models and Naive Bayes

Prof. Dr. Michael Granitzer

Where We Left Off

Lecture 02: 100 students, a count table.

	PML	DB	total
fail	10	6	16
pass	30	54	84
total	40	60	100

$$P(\text{PML} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{PML})\,P(\text{PML})}{P(\text{fail})} = \frac{10}{16}$$

One hypothesis, **one** piece of evidence: Bayes' theorem reads the table backwards.

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Now the hypothesis is a class — fraud or honest — and the evidence a whole transaction. What changes?

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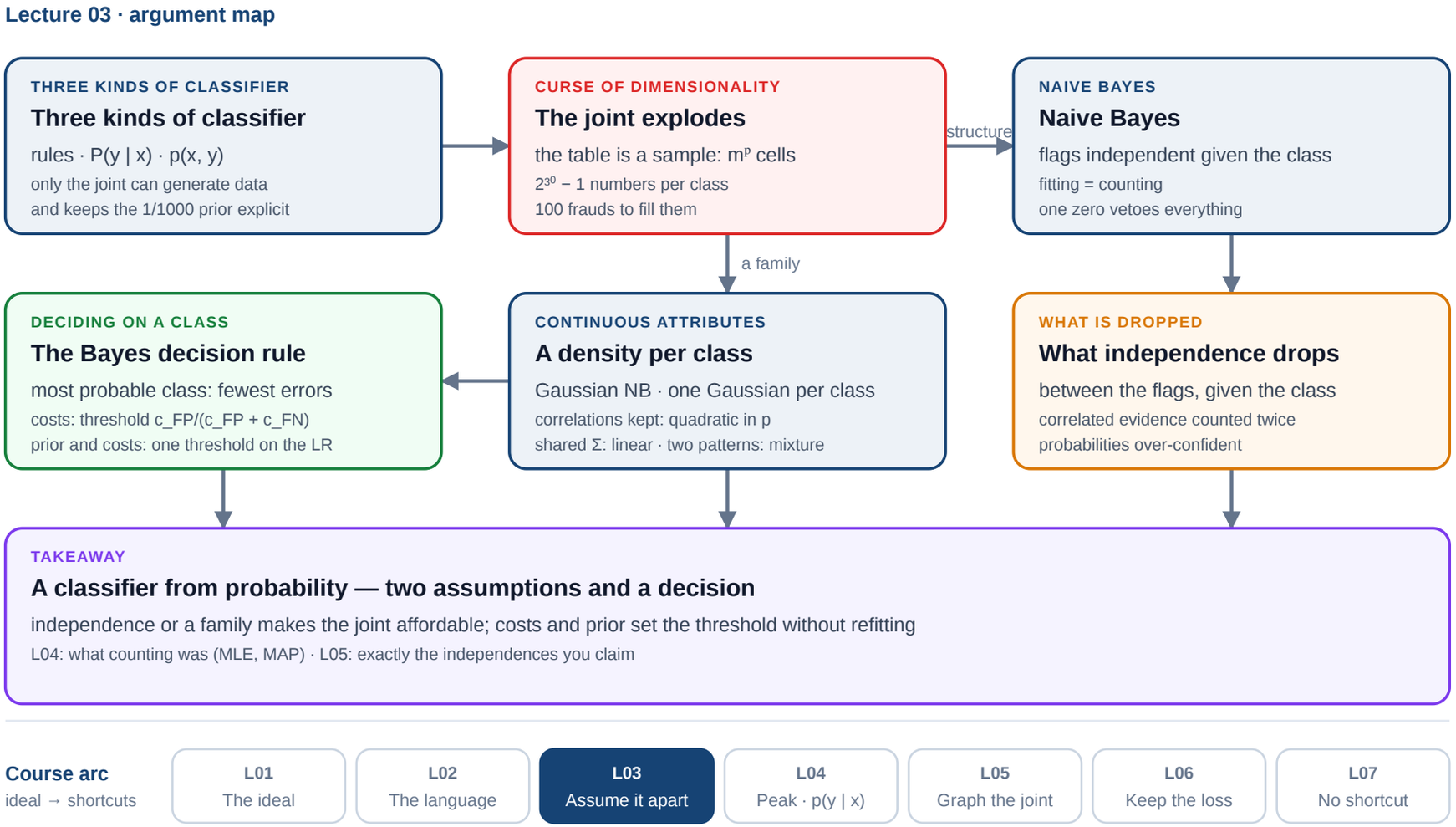
Numbers, not yes/no. What is the probability of *1,250.00 €*?

Probabilities at all? A rule *if amount > 1k € and new device, then fraud* also classifies.

Today: Generative Classifiers and the Bayes Decision Rule

question	module	the answer, in short
Probabilities at all?	**Three Kinds of Classifier**	a rule gives a class; a model of $p(x, y)$ gives $P(y \mid x)$, $p(x)$ and new data — if the joint can be estimated
Many variables at once?	**The Curse of Dimensionality**	the joint explodes; two ways out — simpler structure, a family
Dependent variables?	**Naive Bayes**	assume them independent given the class: one small table each
Numbers, not yes/no?	**Continuous Attributes**	a density per class; one Gaussian keeps the dependences
Which class, then?	**Deciding on a Class**	the most probable one — or a threshold set by costs and prior

Argument Map



Three Kinds of Classifier

Guiding Question: A rule, a model of $P(y \mid x)$, a model of how the data arise — what can each one tell us about a transaction, and what does the most informative one cost?

Data and Hypothesis as Random Variables

amount	distance	…	fraud?
49.90 €	3 km	…	no
1,250 €	900 km	…	**yes**
X_1	X_2	…	Y

- **Data** $x = (x_1, \ldots, x_p)$: the values of the attributes $X_1, \ldots, X_p$ — what we observe.
- **Hypothesis** y: the value of the class variable Y — what we want to know.
- **Decision:** the posterior $P(y \mid x)$.

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The student table had hypothesis A = "PML" and evidence B = "fail". Which of the two is the data, which the hypothesis?

Parameters θ

A **model** is a family of distributions $p(\,\cdot\,;\theta)$. The **parameters** θ pick one member; training sets them.

Read the notation

$p(x \mid y;\theta)$ — the probability of data x, **given** the hypothesis y, **for** the parameter value θ.

The bar conditions on a random variable, the semicolon fixes a parameter.

Example

Fraud share: $P(Y = \text{fraud};\theta) = \pi$, one number.

Log amount per class ~ normal: $\mu_{\text{f}}, \sigma_{\text{f}}$ and $\mu_{\text{h}}, \sigma_{\text{h}}$.

Together: $\theta = (\pi, \mu_{\text{f}}, \sigma_{\text{f}}, \mu_{\text{h}}, \sigma_{\text{h}})$.

Rule-Based Classifier

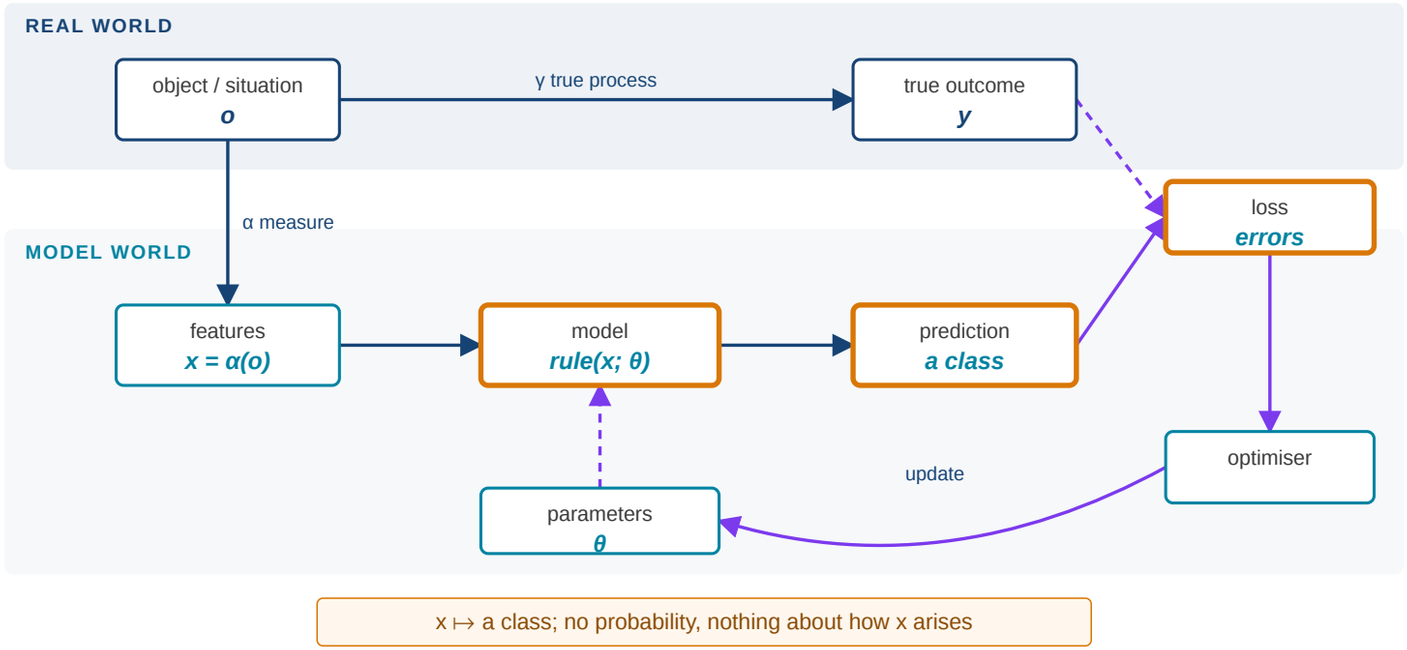
A **rule-based classifier**: rules "if $C_k(x)$ then c_k" plus a default class. The **rules are the parameters** θ.

Example: $x_{\text{amt}} > a$ and $x_{\text{dist}} > b \rightarrow$ fraud, else honest; $\theta = (a, b)$. **Training:** the fewest errors on the n rows,

$$\hat{\theta} = \arg\min_{\theta} \sum_{i=1}^{n} 1[\hat{y}_i \neq y_i]$$

- y_i: label of row i; $\hat{y}_i = \text{rule}(x_i; \theta)$: the rule's class
- $1[\cdot]$: 1 if true, else 0 — the sum counts errors
- $\arg\min_\theta$: the θ with the fewest; $\hat{\theta}$: the fitted θ

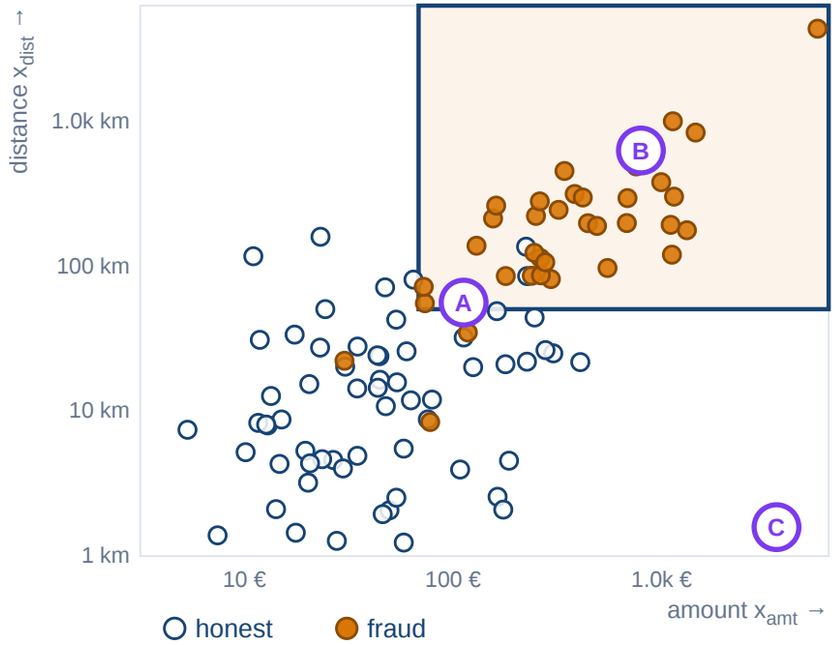
Discriminant function: rules



x ↦ a class; no probability, nothing about how x arises

Rules From Data

A and B get the same answer; C gets one without any warning



Learned rule

if amount > 68 €
and distance > 51 km
then fraud, else honest

training errors: 5 of 98

fraud share in the box: 94%
— the only "score" the rule has

A: fraud, 94%
just inside the box

B: fraud, 94%
deep inside the box

C: honest
3.5k € at 1.6 km — no such row

Which transaction is the rule surest about — A, B or C?

What a Rule Can Not Say

- ✗ **How sure.** One share for the whole box — A and B alike.
- ✗ **Honestly.** The share is measured on the rows that chose the box: optimistic, 100% for a box searched until pure.
- ✗ **Under a new base rate.** The training mix is baked into the thresholds.
- ✗ **That an input is strange.** No model of x: C gets an answer like any other.

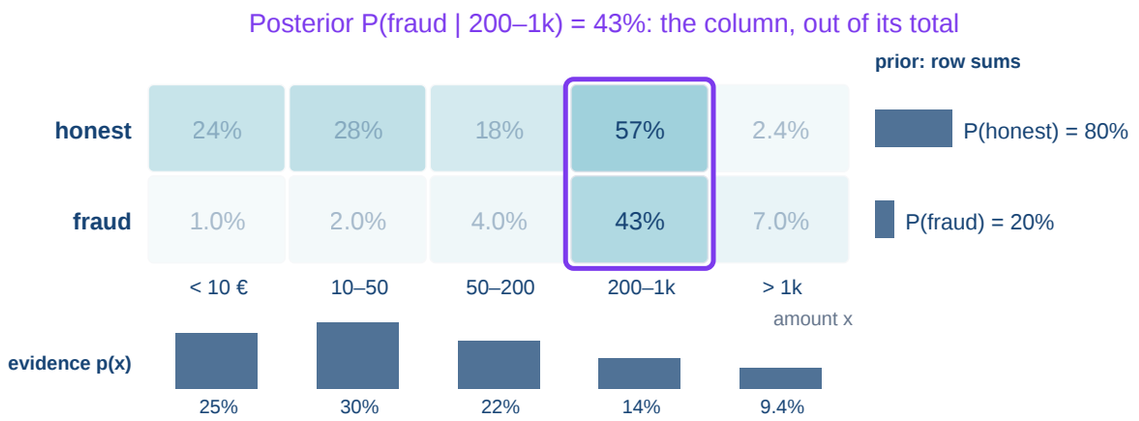
Examples

- **Fraud rule engines:** "amount > 1k € and new device → block".
- **Decision trees:** rules learned by splitting, one class per leaf.

A rule is a **discriminant function**: $x \mapsto$ a class. No distribution is modelled.

The Joint Distribution $p(x_1, \ldots, x_p, y)$

A **probabilistic** classifier starts from the joint distribution of data and hypothesis. Every question is read off it:



question	read off the joint
Is this transaction fraud?	$P(y \mid x) = p(x, y) \,/\, p(x)$
How does fraud look?	$p(x \mid y) = p(x, y) \,/\, P(y)$
How common is fraud?	$P(y) = \sum_x p(x, y)$
How unusual is this transaction?	$p(x) = \sum_y p(x, y)$
The distance is missing — now?	sum x_{dist} out, then condition

> The joint is the course's ideal: model everything, then condition.

Discriminative Classifier

A **discriminative classifier** models only $P(y \mid x; \theta)$ — nothing about which x occur.

Example: logistic regression, 3 numbers:

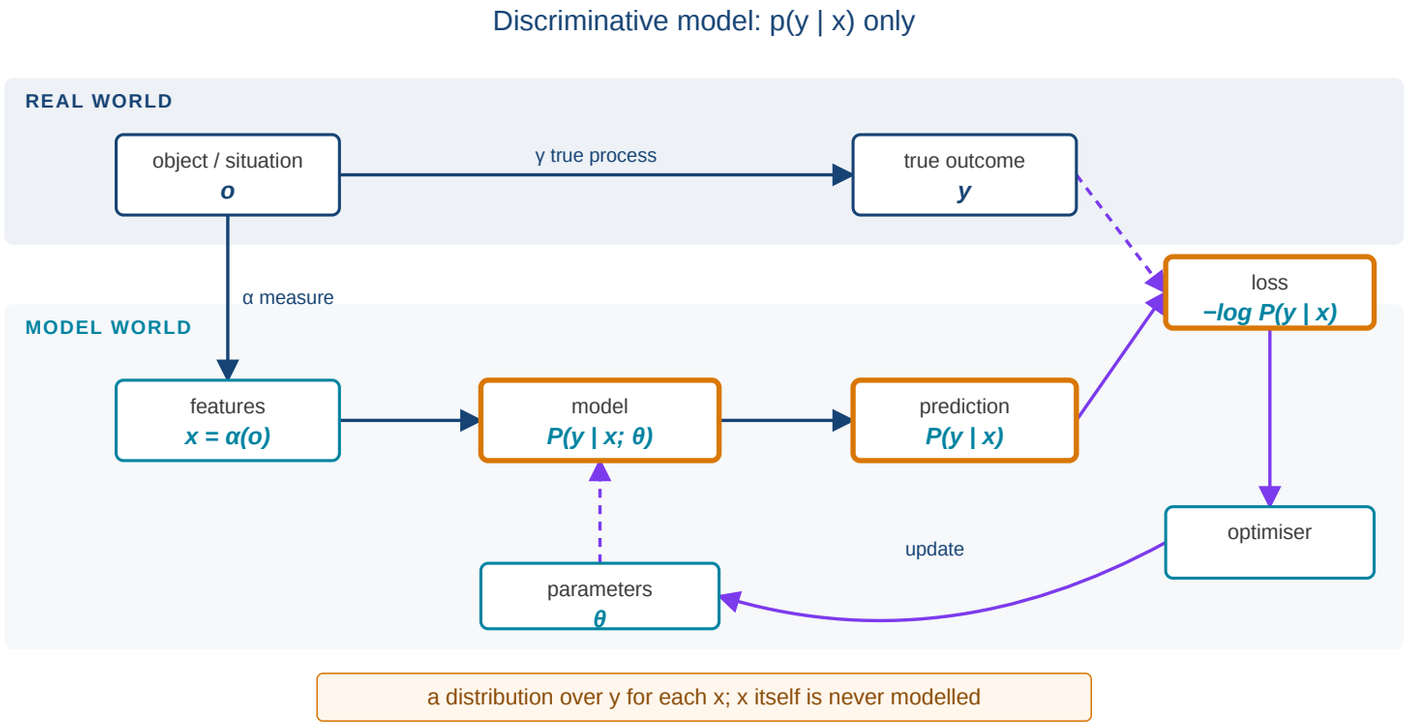
$$z = \theta_0 + \theta_{\text{amt}} \log x_{\text{amt}} + \theta_{\text{dist}} \log x_{\text{dist}}$$

$$P(\text{fraud} \mid x; \theta) = \sigma(z) = 1/(1 + e^{-z})$$

Training: the most probable labels,

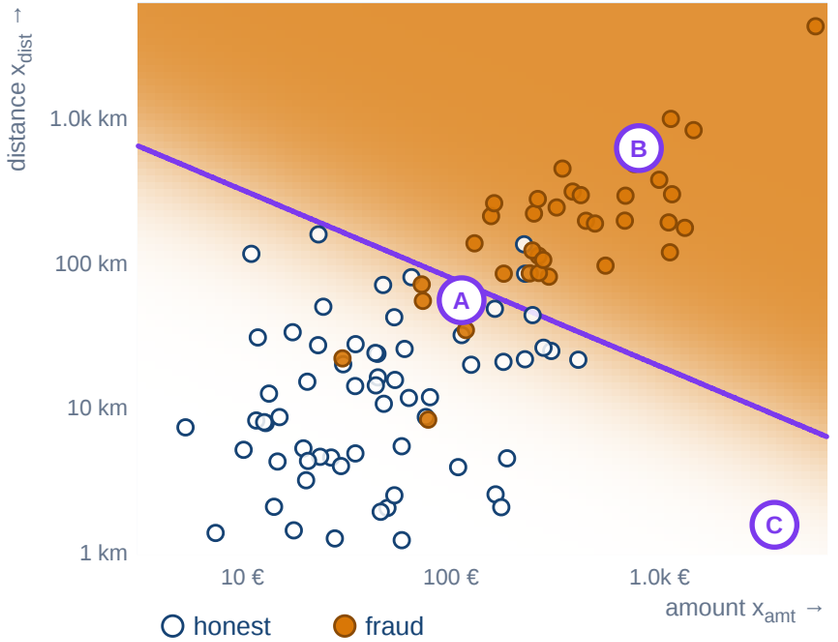
$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \log P(y_i \mid x_i; \theta)$$

x_i stands only right of the bar: given, never modelled.



$P(y \mid x)$ From Data

A and B now differ; C (3.5k € at 1.6 km) still gets an answer, without warning



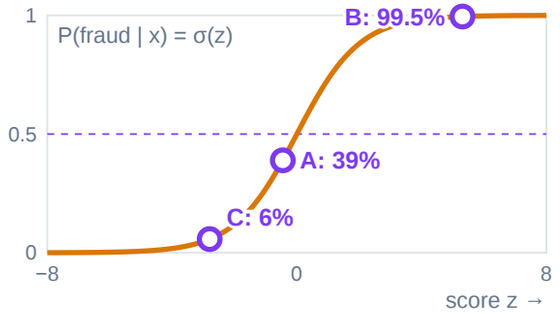
Logistic regression

$z = -11.5 + 2.2 \log x_{amt} + 3.7 \log x_{dist}$

$P(\text{fraud} \mid x) = \sigma(z) = 1 / (1 + e^{-z})$

z: a score, the log-odds — any number

×10 amount: odds ×9 · ×10 distance: odds ×40



7 of 98 training errors · no p(x | y), no p(x)

Compare with the rule: what is new for A and B — and what is still missing for C?

Generative Classifier

A **generative classifier** models how the data arise — $p(x \mid y; \theta)$ and $P(y; \theta)$ — and obtains $P(y \mid x)$ by Bayes' theorem.

Example: a normal per class and the fraud share, 11 numbers:

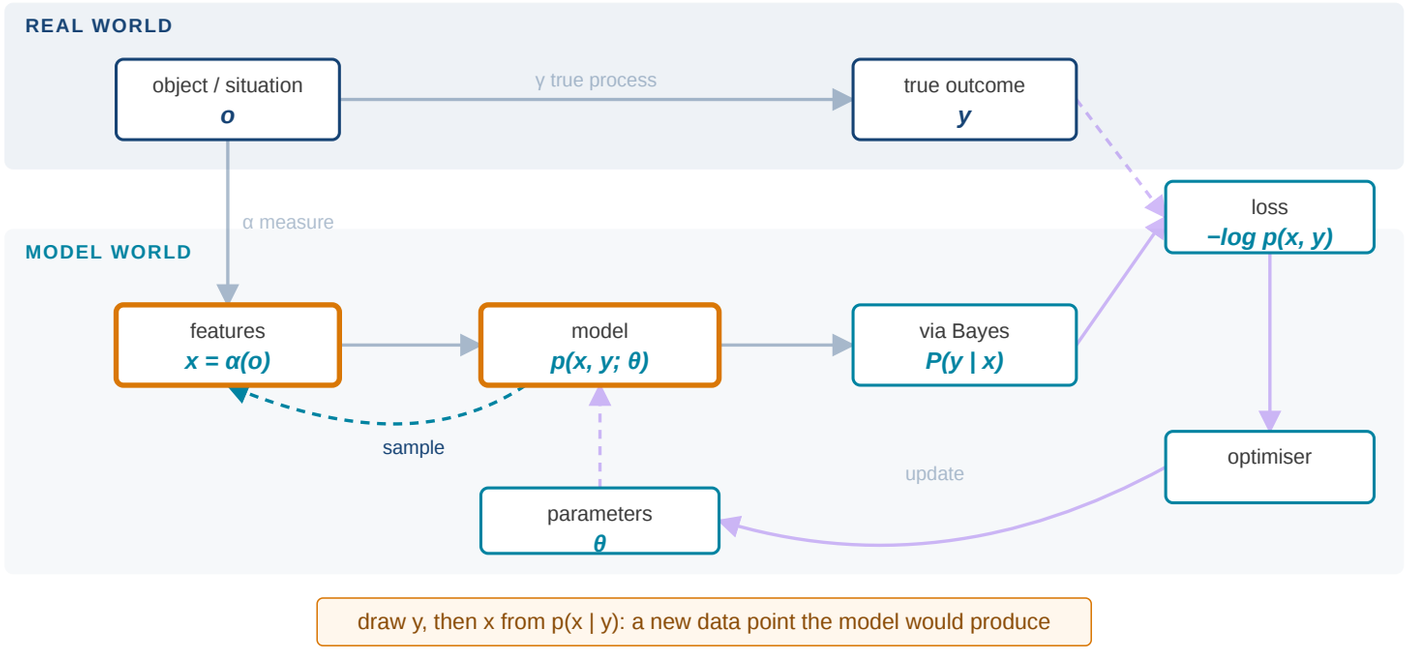
$$p(x \mid y; \theta) = \mathcal{N}\big((\log x_{\text{amt}}, \log x_{\text{dist}});\ \mu_y, \Sigma_y\big)$$
$$P(\text{fraud}; \theta) = \pi$$

Training: the most probable rows,

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{n} \big[\log p(x_i \mid y_i; \theta) + \log P(y_i; \theta)\big]$$

x_i is scored too: modelled, not only given.

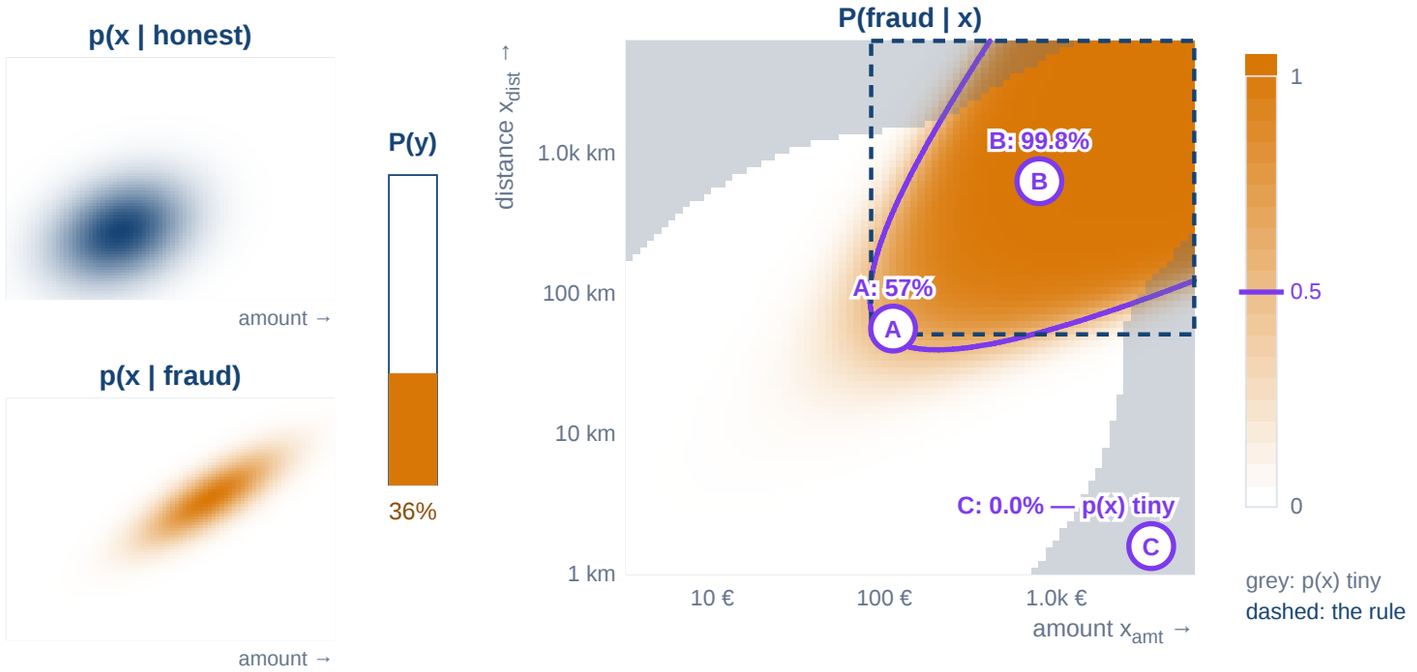
Generative model, sampling



draw y, then x from p(x | y): a new data point the model would produce

$p(x \mid y)$ From Data

Graded, prior-aware — and it knows where it has seen nothing



Set $P(\text{fraud})$ to 0.1%. What happens to the boundary — and why can the rule not do this?

Generating Data

Generating is sampling from the model: draw $y \sim P(y)$, then $x \sim p(x \mid y)$ — here y = spam, x = the words of a mail.

Naive Bayes

each word from p(word | spam)

*“won now won to click click a **now** …”*

Language model (toy)

each word from p(word | the words before)

“you have won a free prize now.”

The vocabulary: 13 words, the same for both models



Which mail reads like a real one — and what does naive Bayes leave out?

Three Kinds of Classifier

	rule	discriminative	generative
models	$x \mapsto$ class	$P(y \mid x; \theta)$	$p(x \mid y; \theta)\, P(y; \theta)$
fitted here	2 thresholds	3 weights	11 numbers
A, just inside the box	fraud, 94%	39%	57%
B, deep inside	fraud, 94%	99.5%	99.8%
C, unlike any row	honest, **no warning**	6%, **no warning**	0.0%, **p(x) tiny**
new base rate	✗ refit	correct from outside	✓ swap $P(y)$
missing field	✗ ad hoc	✗ ad hoc	✓ sum it out
generate data	✗	✗	✓
examples	decision tree	logistic regression	naive Bayes, language model

The price of the right column: a distribution over **all** attributes, per class.

The Curse of Dimensionality

Guiding Question: A generative classifier needs $p(x \mid y)$ over all attributes at once. How many numbers is that — and how can a model get by with fewer?

Bayes' Theorem With Two Red Flags

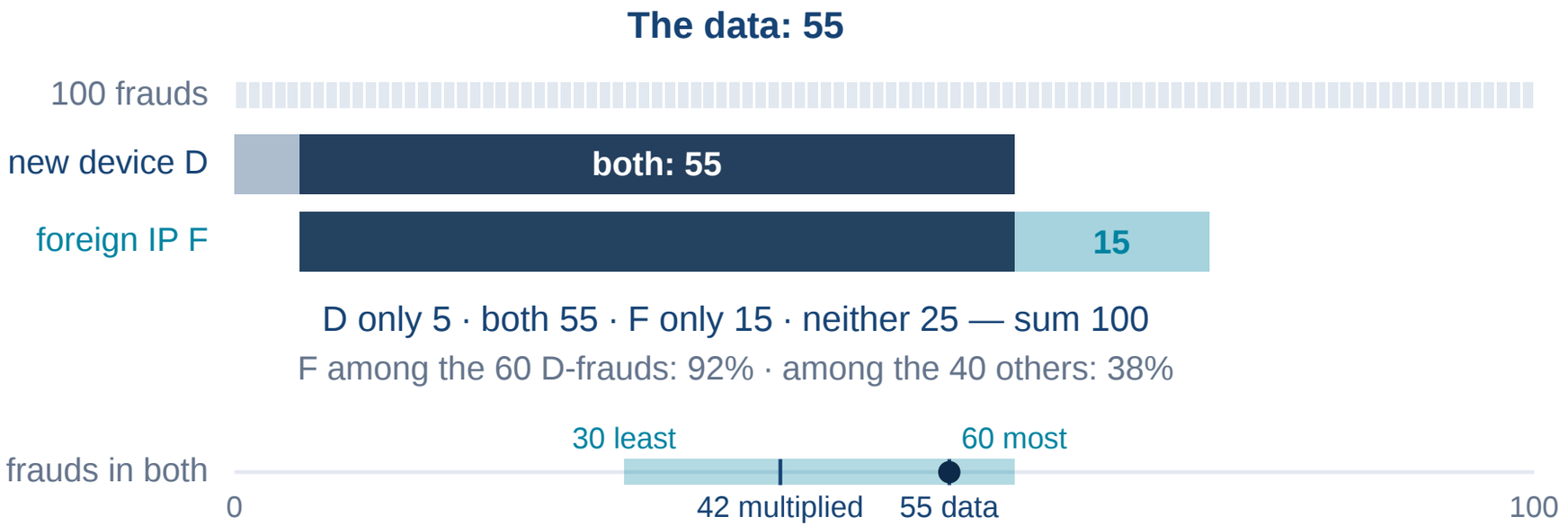
Two flags at once: D = new device, F = foreign IP.

$$P(\text{fraud} \mid D \cap F) = \frac{P(D \cap F \mid \text{fraud}) \cdot P(\text{fraud})}{\sum_y P(D \cap F \mid y)\, P(y)}$$

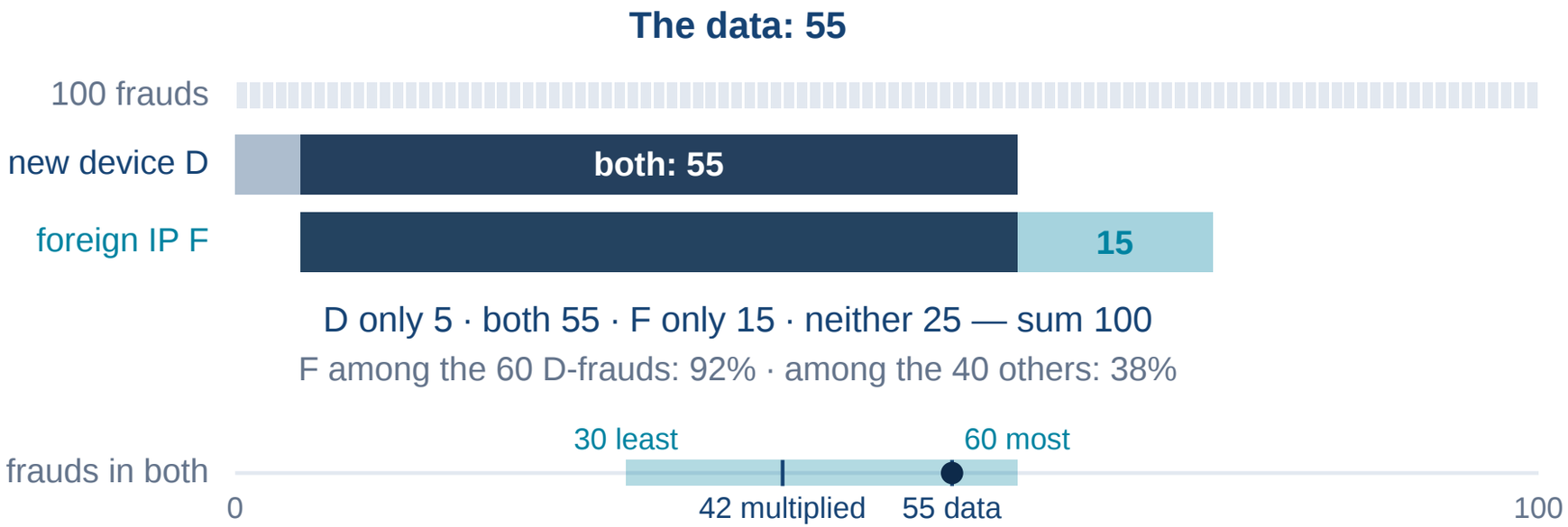
Nothing in the theorem changes. New is the evidence: a **conjunction** $D \cap F$, whose likelihood is needed for every class y.

Among 100 frauds, 60 show D and 70 show F. How many show both?

The Joint of Two Flags



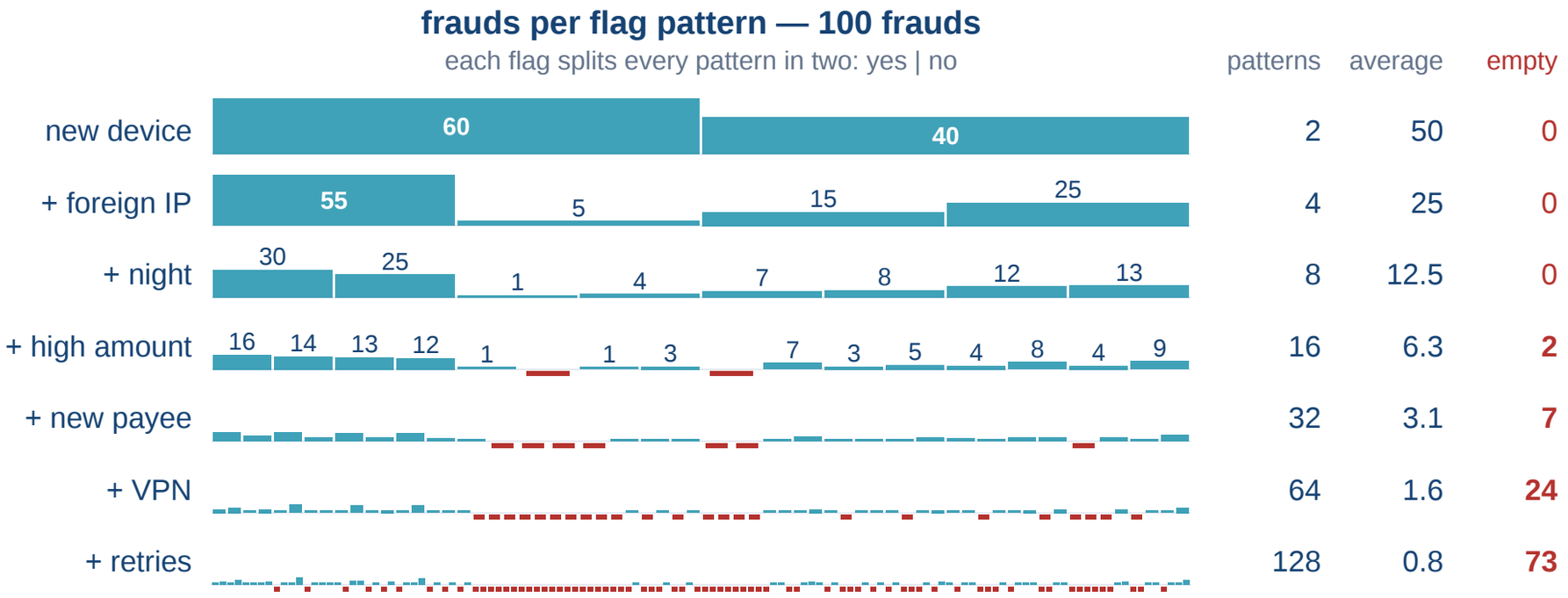
The Joint of Two Flags



Not fixed by the parts: 30 to 60 all fit both rates, so $P(D \cap F \mid y)$ is a number of its own. The product $0.60 \cdot 0.70 = 0.42$ assumes D and F **independent given the class** — naive Bayes.

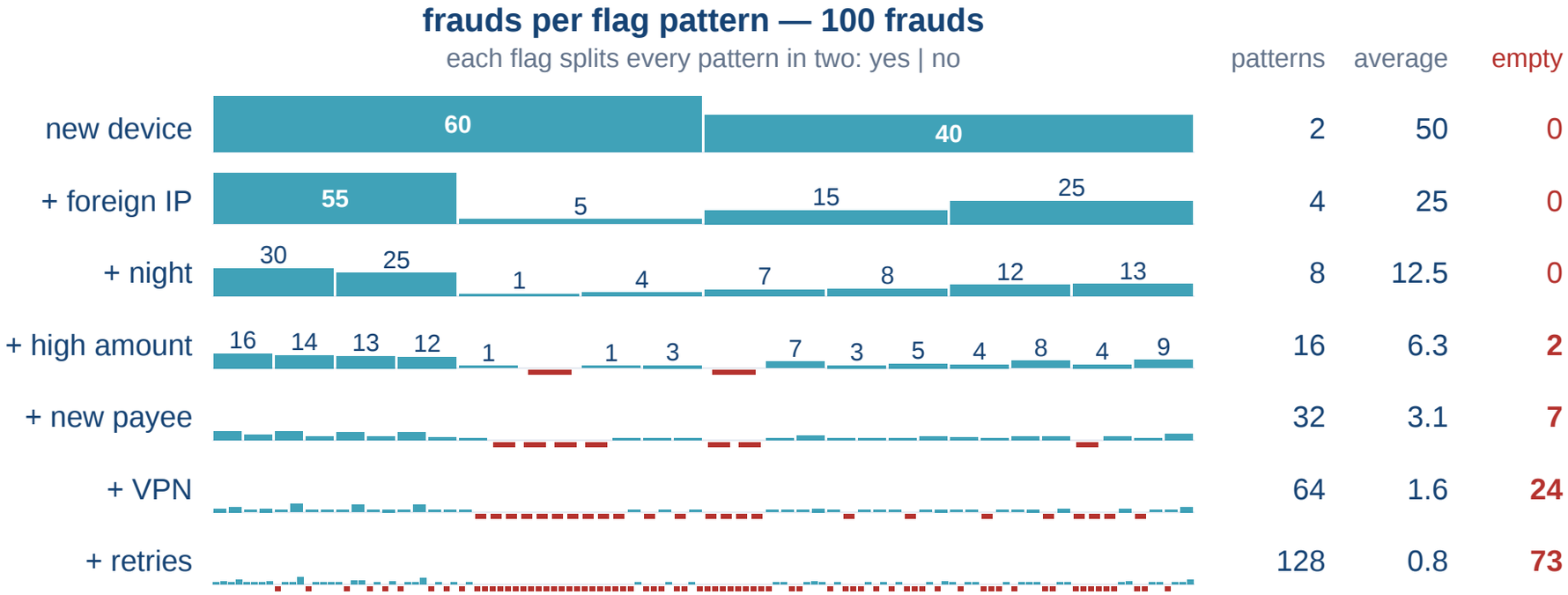
One Number per Flag Pattern

A third flag, *night*: how many groups now — and how many frauds in each?



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Every flag **doubles** the patterns — p flags, 2^p patterns, $2^p - 1$ numbers per class — and **halves** the frauds behind each: at 7 flags, 73 of 128 patterns are **empty**.

The Training Table Is a Sample

amount (€)	country	hour	new device?	… 30 flags	fraud?
49.90	DE	14	no	…	no
1,250.00	BR	3	yes	…	**yes**
12.50	DE	19	no	…	no

Each column is a **random variable** X_j, each row one **draw** from $p(x, y)$. The fraud rows are draws from $p(x \mid \text{fraud})$.

Estimate $p(x \mid \text{fraud})$ by counting: of 100 frauds, how many look exactly like the next transaction?

The Training Table Is a Sample

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> Almost always **none**. Counting needs examples of every combination of values.

The Joint Explodes

30
yes/no red flags per transaction

→

$2^{30} - 1$
≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100
frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

The Curse of Dimensionality

attributes	value combinations (10 values each)	frauds per combination (100 frauds)
1	10	10
2	100	1
3	1,000	0.1
30	10^{30}	10^{-28}

Every attribute **multiplies** the cells and **divides** the data per cell. Counting needs data in every cell — a full joint over many attributes can never be estimated by counting.

Two Ways Out

① Simplify the structure

Assumption on the data: attributes **independent given the class** — one small table each, not one huge joint.

30 flags: **30** numbers per class, not $2^{30} - 1$.

NAIVE BAYES — TODAY; GRAPHS — LECTURE 05

② Use a parametric family

Assumption on the model: a **formula with few parameters** instead of a table of counts.

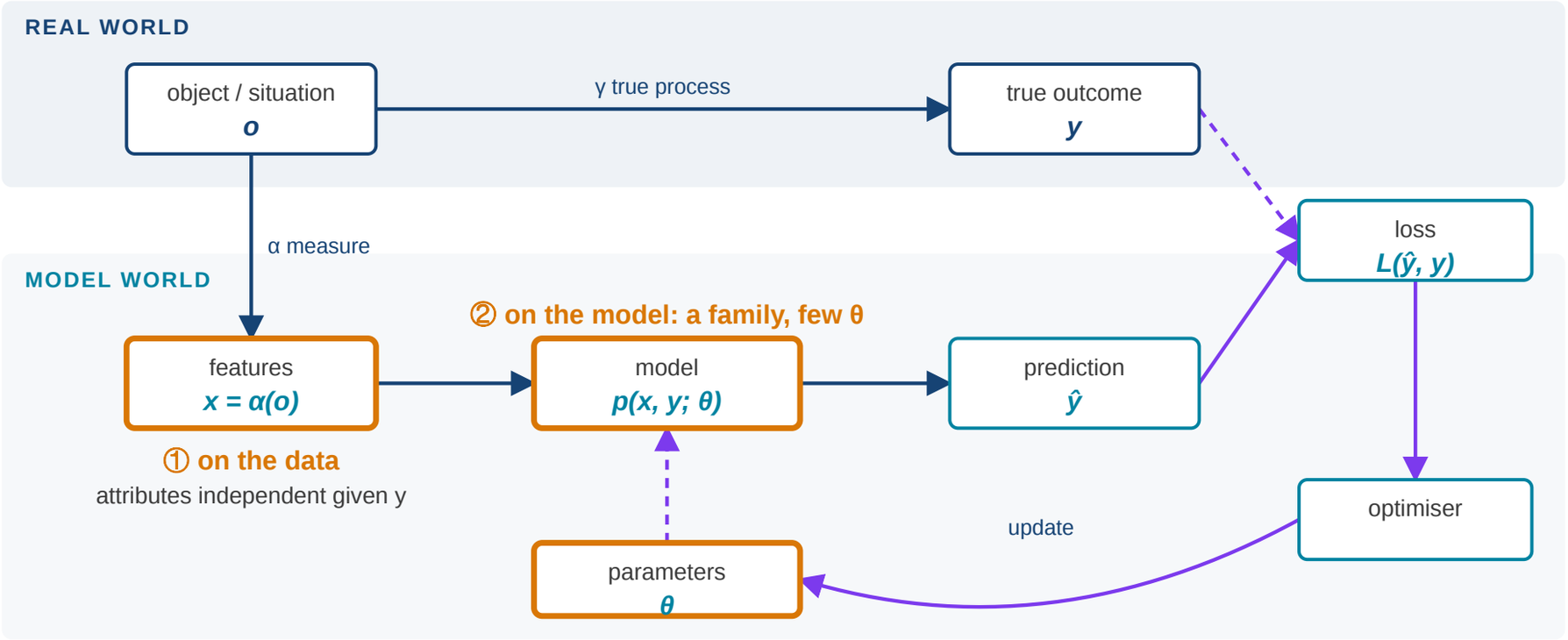
amount given fraud ~ log-normal(μ, σ): **2** numbers, not 100 bins.

LECTURE 02; A NORMAL PER CLASS — TODAY

Both are **assumptions**. They make the model estimable — and wrong wherever the world disagrees with them.

Assumptions on the Data and on the Model

② Use a parametric family: an assumption on the model



which distributions the model may use: a family — it fixes how many θ are fitted

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Mini-Exercise

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15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. Counting that joint needs data in every combination of values, which **explodes** with the number of attributes. The way out is an assumption: **simpler structure**, **parametric families**, or both.

Naive Bayes

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

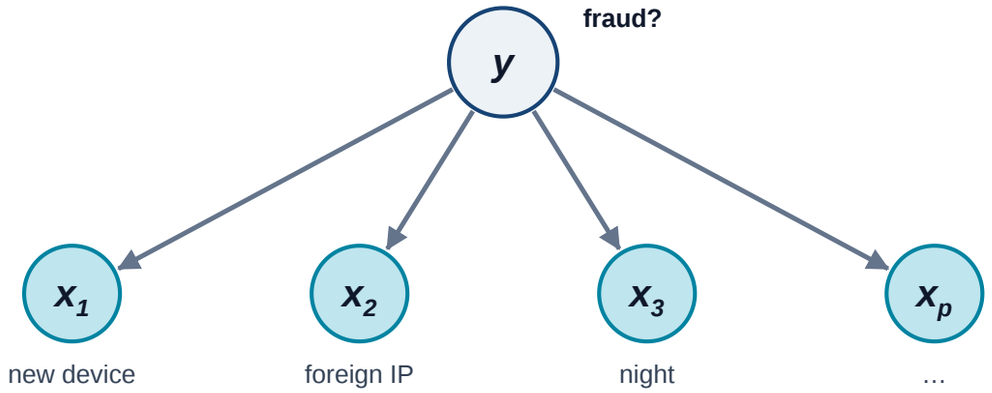
Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

Among frauds only: does a new device make a foreign IP more likely?

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

Assume the attributes are **independent given the class**:

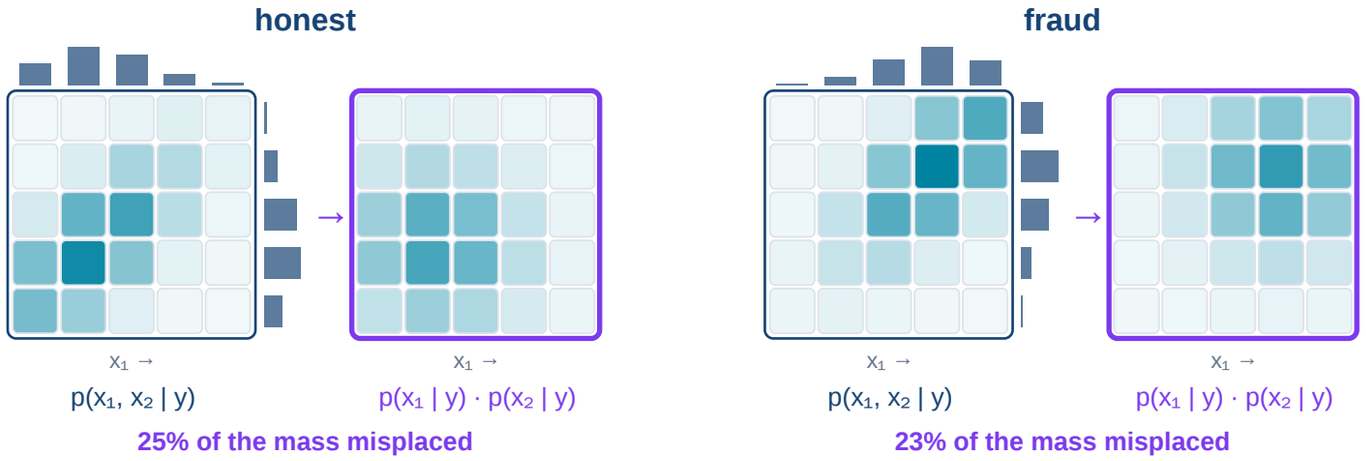
$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 1 — assume it apart. Dropped: every dependence *within* a class.

Assuming It Apart, Drawn

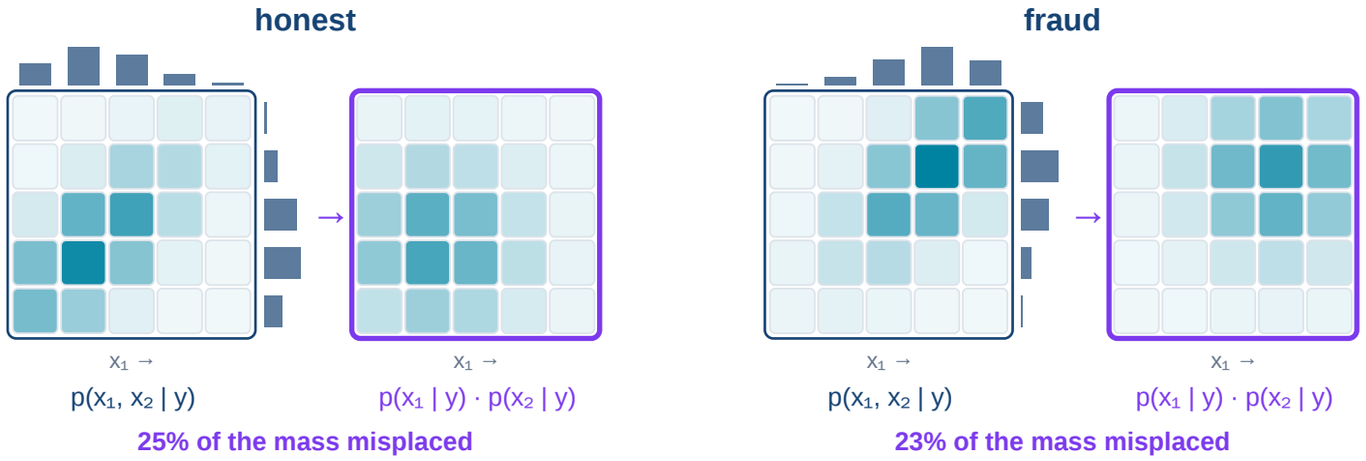
The tilt within each class is what naive Bayes drops



Predict: set the correlation to 0. What happens to the two pictures?

Assuming It Apart, Drawn

The tilt within each class is what naive Bayes drops



Predict: set the correlation to 0. What happens to the two pictures?

They become **identical**. Naive Bayes is exact exactly when the attributes are independent **within** each class.

A Distribution per Attribute and Class

Naive Bayes needs $p(x_j \mid \text{fraud})$ and $p(x_j \mid \text{honest})$ for each attribute on its own. How would you estimate them for these four?

A Distribution per Attribute and Class

Naive Bayes needs $p(x_j \mid \text{fraud})$ and $p(x_j \mid \text{honest})$ for each attribute on its own. How would you estimate them for these four?

attribute x_j	values	model for $p(x_j \mid y)$	fitted from the class's rows by	numbers per class
new device	yes / no	Bernoulli(θ_y)	the share of *yes*	1
country	k countries	categorical	the share of each country	$k - 1$
logins, last hour	0, 1, 2, …	a **table** of the values — or **Poisson** (λ_y)	the share of each value — or the **mean** count	one per value — or 1
amount	positive, skewed	**normal** on the log amount	mean and spread of $\log x$	2

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Naive Bayes fixes the **structure** (way out ①). Each factor still needs a model: a **table**, fitted by counting, or a **family** with few parameters (way out ②). Both are fitted per class, in one pass over the rows.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

One small table per attribute

For flags and categories, each factor is a **table**: a categorical — for a yes/no flag, a Bernoulli.

Estimated by its share

$P(\text{new device} \mid \text{fraud})$: of the 100 frauds, how many came from a new device? 60, so 0.6.

Naive Bayes over tables is fitted in **one pass**, by counting — no optimiser. Why a share is the right estimate — and when it is not — is Lecture 04.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Likelihood Ratios From Counts

The training table: 100,000 transactions, **100** fraud and **99,900** honest.

A new device shows up in 60 frauds — and in 9,990 honest transactions. How strong is it as evidence for fraud?

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flag	frauds with it	$P(\text{flag} \mid \text{fraud})$	honest with it	$P(\text{flag} \mid \text{honest})$	likelihood ratio
new device	60 of 100	0.60	9,990 of 99,900	0.10	0.60 / 0.10 = **6**
foreign IP	70 of 100	0.70	4,995 of 99,900	0.05	0.70 / 0.05 = **14**
night	50 of 100	0.50	19,980 of 99,900	0.20	0.50 / 0.20 = **2.5**

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> Strength is a ratio of **shares**, not of counts: a new device is **6 times as common** among frauds — although 9,990 of its 10,050 transactions are honest.

Three Red Flags

Prior odds $100 : 99{,}900 = 1 : 999$. Likelihood ratios: new device **6**, foreign IP **14**, night **2.5**.

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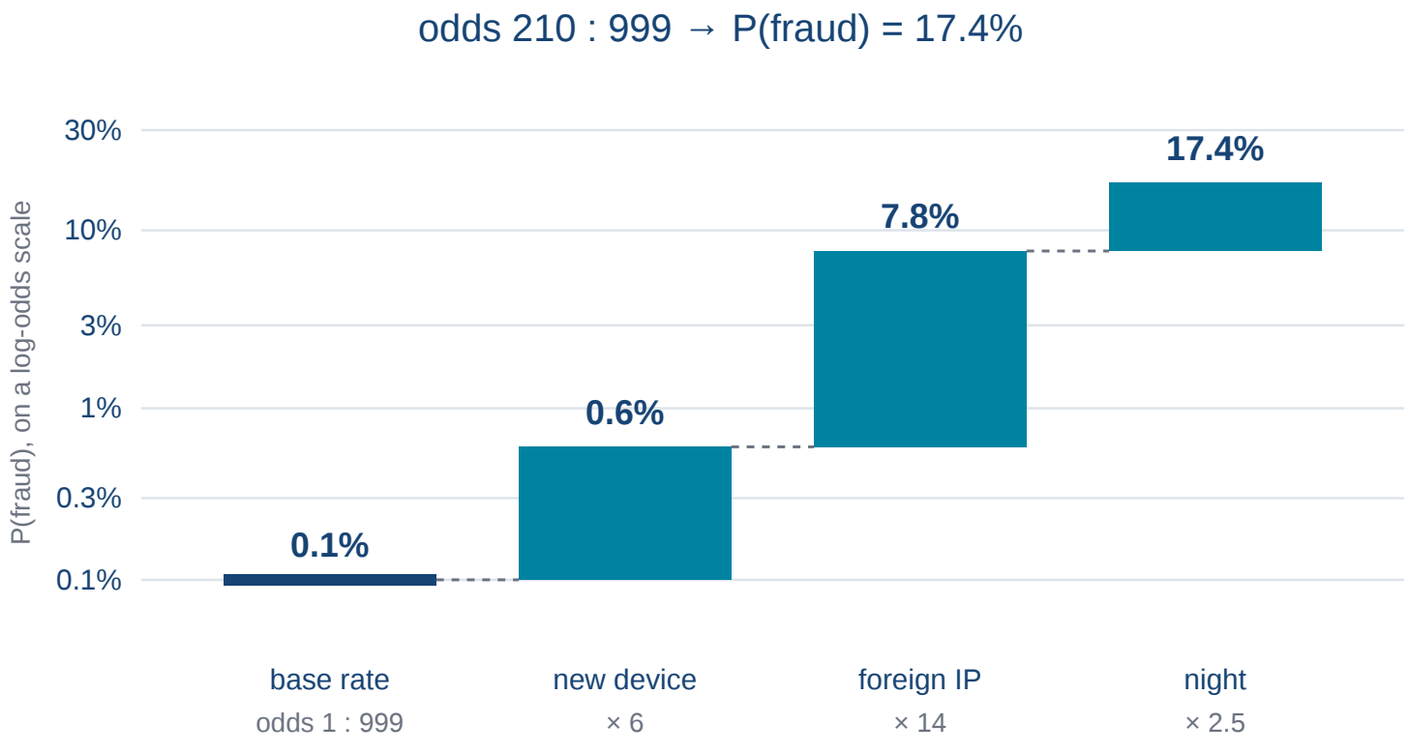
$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up

Take the logarithm of the odds form: the **product** of ratios becomes a **sum**.

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = \log \frac{P(\text{fraud})}{P(\text{honest})} + \sum_j \log \text{LR}_j$$



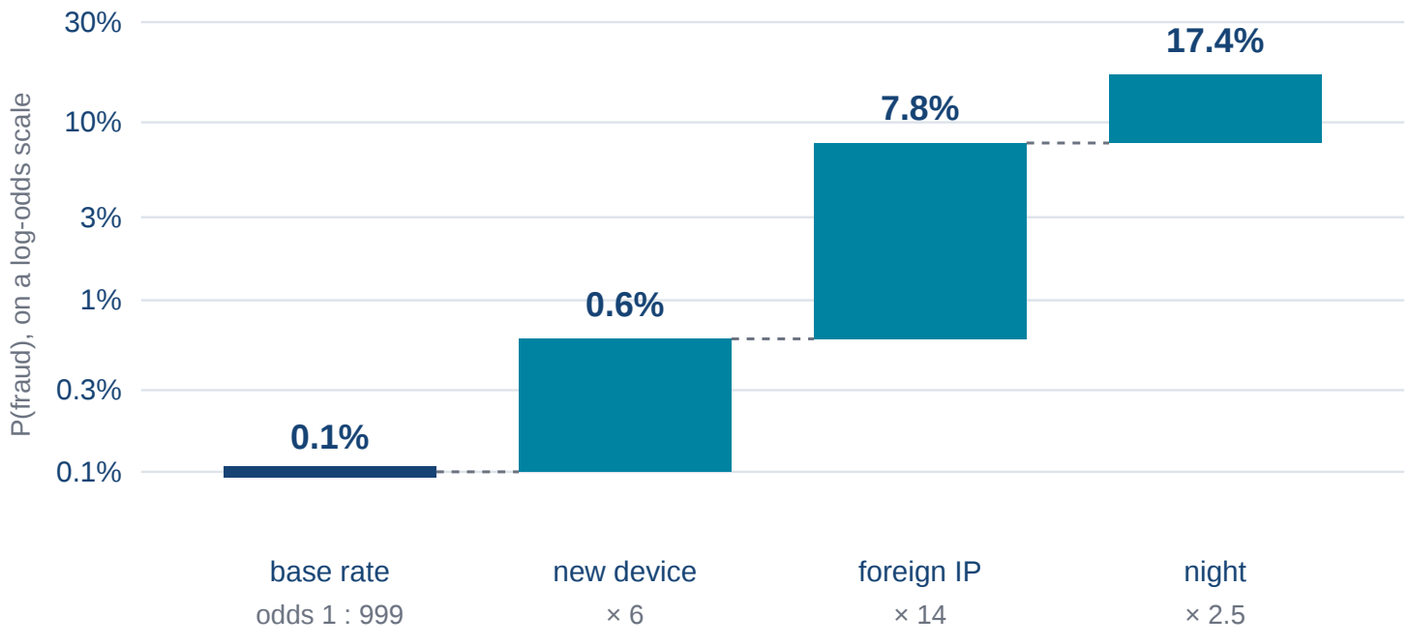
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odds 210 : 999 → P(fraud) = 17.4%



Predict before each click: which flag moves the probability most?

evidence	odds	log-odds	P
base rate	$1:999$	-6.91	0.1%
new device	$\times 6$	$+1.79$	0.6%
foreign IP	$\times 14$	$+2.64$	7.8%
night	$\times 2.5$	$+0.92$	17.4%
sum	$210:999$	$\mathbf{-1.56}$	**17.4%**

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

COUNTING, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — a share of a small count, read as certainty.

Laplace Smoothing

Laplace smoothing pretends each of the k_j values was seen **once more**:

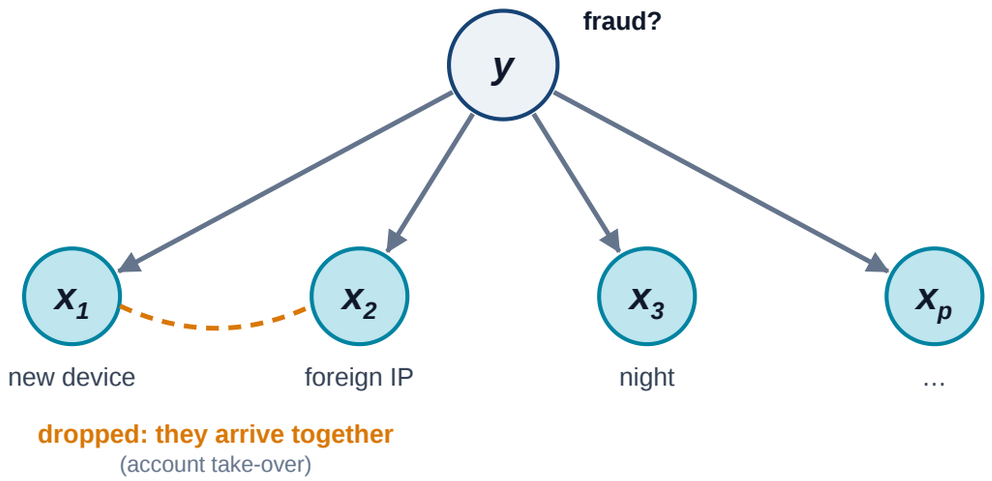
$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$.

count among 100 frauds	share	smoothed
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

> Where the count is large, nothing changes; where it is 0, the veto is gone. Why **+1**, and not +5? Lecture 04: smoothing is a **prior**.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

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Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

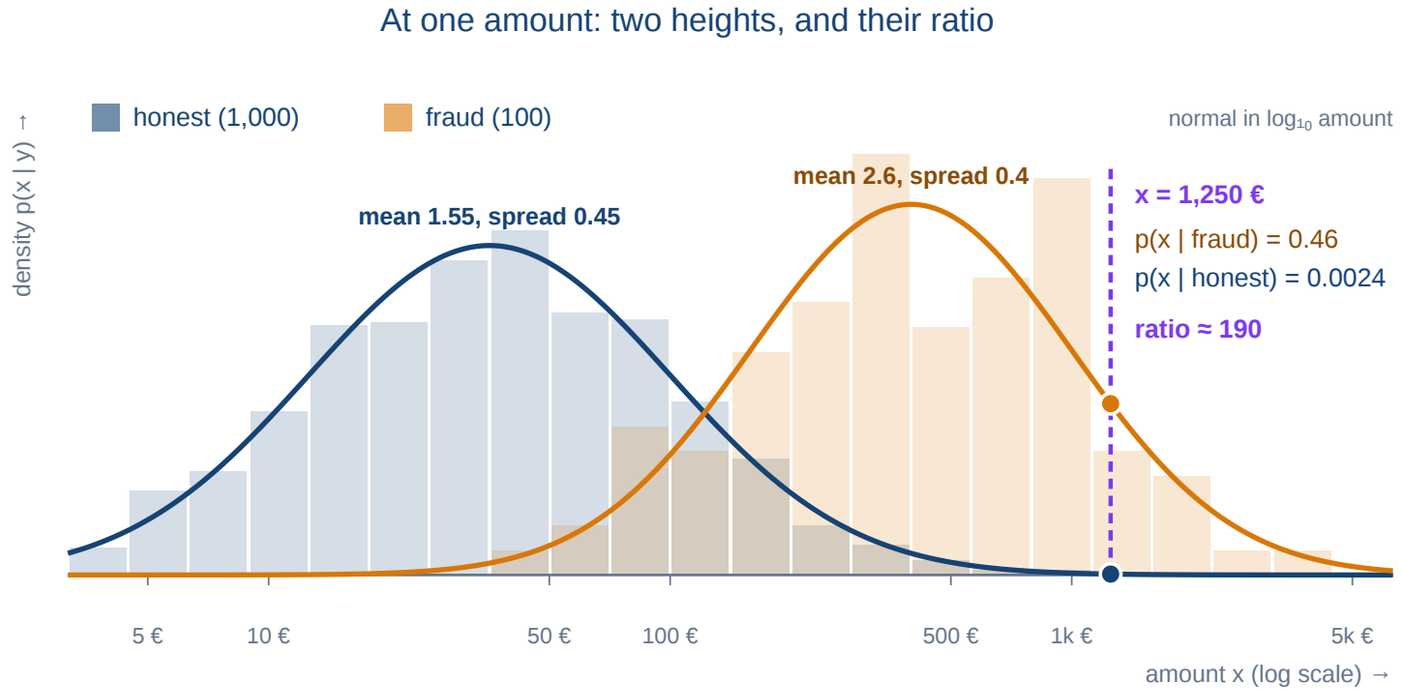
Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and fits every factor by **counting**, smoothed by pseudo-counts so no zero can veto. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Continuous Attributes

Guiding Question: An amount of 1,250 € has probability zero. How does a number enter Bayes' theorem — and what does a model of many numbers cost?

A Density per Class



What is the probability that a fraud is exactly 1,250.00 €?

Bayes' Theorem With Densities

For a continuous attribute, $p(x \mid y; \theta)$ is a **density** — probability per unit of x. Bayes' theorem holds with densities:

$$P(y \mid x) = \frac{p(x \mid y; \theta)\, P(y)}{\sum_{y'} p(x \mid y'; \theta)\, P(y')}$$

Why it works. An interval $[x, x + \Delta]$ has probability $\approx p(x \mid y)\, \Delta$ in each class; the Δ cancels.

Normal per class, $\theta_y = (\mu_y, \sigma_y)$:

$$p(x \mid y; \theta) = \frac{1}{\sigma_y \sqrt{2\pi}}\, e^{-(x-\mu_y)^2 / 2\sigma_y^2}$$

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At 1,250 €: likelihood ratio $0.46 / 0.0024 \approx 190$. With prior odds $1 : 999$, $P(\text{fraud} \mid x) = 190 / 1{,}189 \approx 16\%$.

Gaussian Naive Bayes

Naive Bayes with a **family per attribute** — a normal for each number, a Bernoulli for each flag — all independent given the class:

$$p(x \mid y;\theta) = \prod_{\text{numbers } j} \mathcal{N}(x_j;\ \mu_{jy}, \sigma^2_{jy}) \cdot \prod_{\text{flags } j} \theta_{jy}^{x_j}(1-\theta_{jy})^{1-x_j}$$

attribute of the transaction	fitted per class	likelihood ratio
amount 1,250 €	mean and spread of log amount	≈ 190
new device: yes	share of rows with the flag	6
both		$190 \cdot 6 = 1{,}140$

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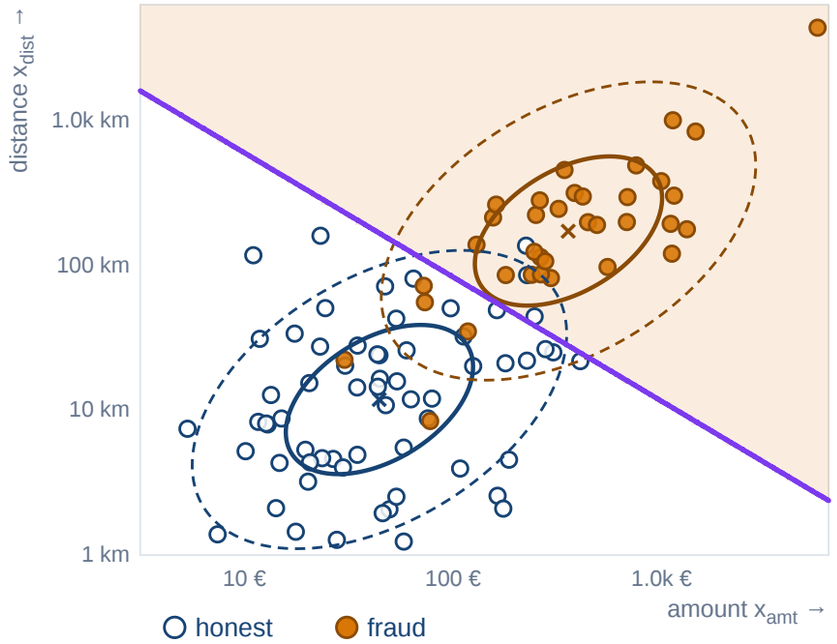
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attribute of the transaction	fitted per class	likelihood ratio
amount 1,250 €	mean and spread of log amount	≈ 190
new device: yes	share of rows with the flag	6
both		$190 \cdot 6 = 1{,}140$

> Posterior odds $1{,}140 : 999$, so **53%** fraud. Fitting is still one pass over the table: a mean and a spread per number and class, a share per flag and class.

One Gaussian per Class

One shared covariance: the boundary becomes straight



One Σ shared by both classes

same spreads and correlation
for honest and fraud

per class: 2 means; Σ once: 3
boundary: a straight line

prior: the training share, 36% fraud

× mean — 1σ - - 2σ
violet: P(fraud | x) = 0.5

In fraud, amount and distance rise together. Which ellipses can follow the tilt — and what does one covariance for both classes do to the boundary?

The Multivariate Normal

$$\mathcal{N}(x;\ \mu, \Sigma) = \frac{1}{(2\pi)^{p/2}\sqrt{\det \Sigma}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big)$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

The **covariance matrix** Σ: spreads on the diagonal, covariances off it.

model of $p(x \mid y)$ for p numbers	numbers per class	$p = 30$
counts, 10 bins per attribute	$10^p - 1$	10^{30}
Gaussian naive Bayes	$2p$	60
one Gaussian per class	$p + p(p+1)/2$	495

The family alone defeats the explosion: **quadratic** in p, not exponential — and the correlations stay in the model.

A Shared Covariance Gives a Linear Boundary

With the **same** Σ in both classes, the quadratic terms cancel in the log-odds:

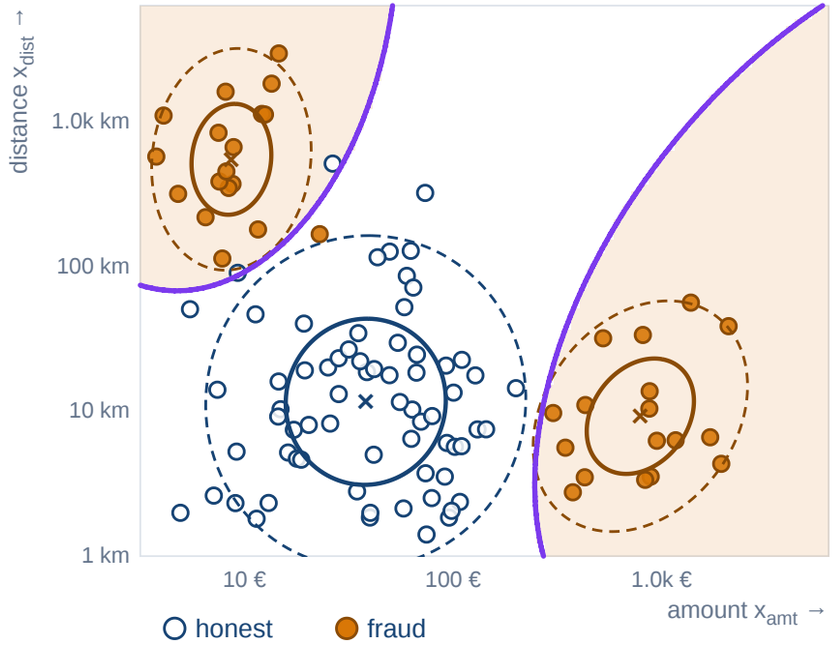
$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = \underbrace{(\mu_\text{f} - \mu_\text{h})^\top \Sigma^{-1}}_{w^\top} x + b \qquad \Longrightarrow \qquad P(\text{fraud} \mid x) = \frac{1}{1 + e^{-(w^\top x + b)}}$$

- **One Σ per class:** a quadratic boundary — *quadratic discriminant analysis*.
- **Shared Σ:** a straight boundary — *linear discriminant analysis*.
- **Diagonal Σ per class:** Gaussian naive Bayes, quadratic as well.

> A logistic function of a linear score: the form that logistic regression (Lecture 04) fits **directly**, without modelling x.

When One Gaussian Is Not Enough

A mixture of two normals for fraud covers both patterns



Fraud: a mixture of two Gaussians

$p(x \mid \text{fraud}) = \pi_1 \, N(\mu_1, \Sigma_1) + \pi_2 \, N(\mu_2, \Sigma_2)$

$\pi = 0.51, 0.49$

fitted here with each pattern known;
in real data it is hidden

prior: the training share, 34% fraud

× mean — 1σ - - 2σ
violet: $P(\text{fraud} \mid x) = 0.5$

Fraud comes in two patterns. Where does a single normal put its mean — and what does that do to the boundary?

Mini-Exercise

Ten numeric attributes, two classes.

How many numbers does Gaussian naive Bayes need — and one full Gaussian per class? Count the prior too.

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41 and 131. Naive: $10 + 10 = 20$ per class, $2 \cdot 20 + 1$. Full: $10 + 55 = 65$ per class, $2 \cdot 65 + 1$.

Takeaway

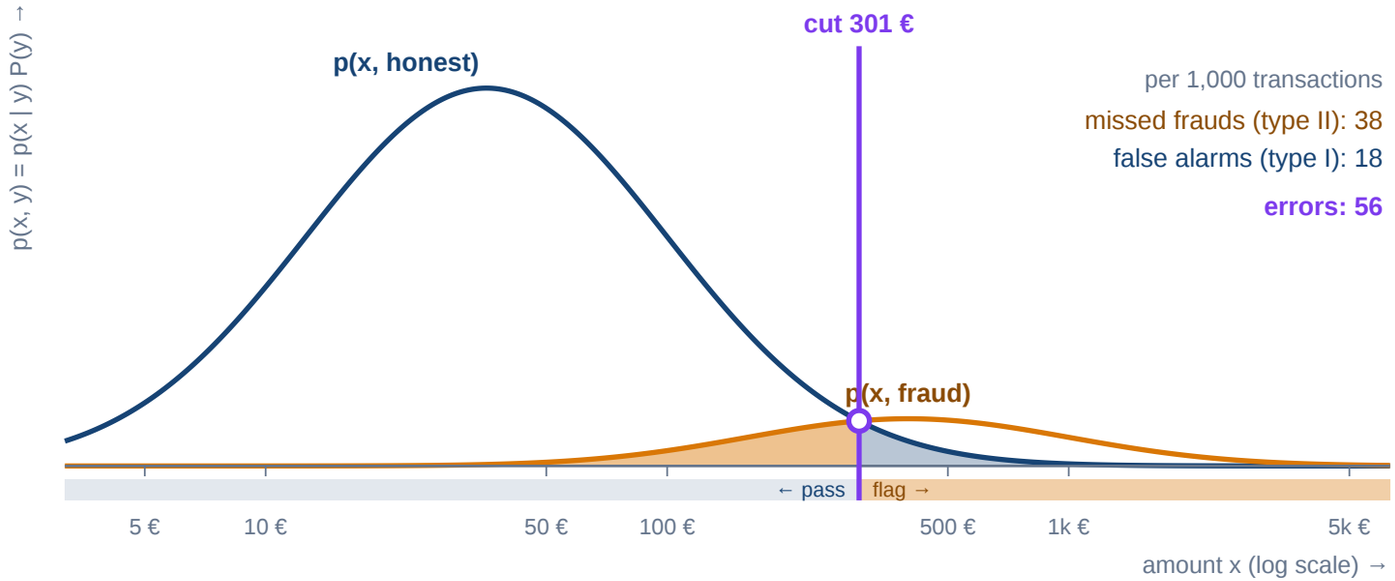
A number enters through a **density** per class; Bayes' theorem needs only the ratio of the two densities. **Gaussian naive Bayes** fits a mean and a spread per attribute and class; **one Gaussian per class** keeps the correlations at a quadratic, not exponential, cost; a **shared covariance** makes the boundary linear. One Gaussian cannot follow a class with two patterns — a **mixture** can.

Deciding on a Class

Guiding Question: The model says 17% fraud. Block, check or let it through — and who decides where the line is?

Decision Regions and Their Errors

The lowest error: cut where the curves cross — P(fraud | x) = 0.5



Flag every transaction above the cut. Where should the cut go to make the fewest errors?

Two Kinds of Error

The best cut of the last slide, about 300 €, per 1,000 transactions (100 of them fraud):

	is fraud	is honest
flag (decide fraud)	62 — true positive	**18 — false positive**: type I error, a false alarm
pass (decide honest)	**38 — false negative**: type II error, a missed fraud	882 — true negative

Move the cut to 1,000 €. What happens to the two errors?

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Move the cut to 1,000 €. What happens to the two errors?

> False alarms fall to 0.6, missed frauds rise to 84: moving the cut **trades one error for the other**. The Bayes rule minimises their **sum**; costs, next, weigh them.

The Bayes Decision Rule

The **Bayes decision rule** decides the most probable class:

$$\hat{y}(x) = \arg\max_y P(y \mid x) = \arg\max_y p(x \mid y)\, P(y)$$

No other rule on the same attributes makes fewer errors.

Why it errs least

At each x, deciding $\hat{y}$ is wrong with probability $1 - P(\hat{y} \mid x)$ — smallest for the most probable class. Best at every x is best on average.

Two classes, as odds

Decide fraud iff $P(\text{fraud} \mid x) > 0.5$, i.e.

$$\text{LR}(x) > \frac{P(\text{honest})}{P(\text{fraud})} = 999$$

Unequal Costs

	is fraud	is honest
check it (a text to the customer)	0	€5 (c_{FP}, false positive)
let it through	€500 (c_{FN}, false negative)	0

With $p = P(\text{fraud} \mid x)$, at which p is checking the cheaper action?

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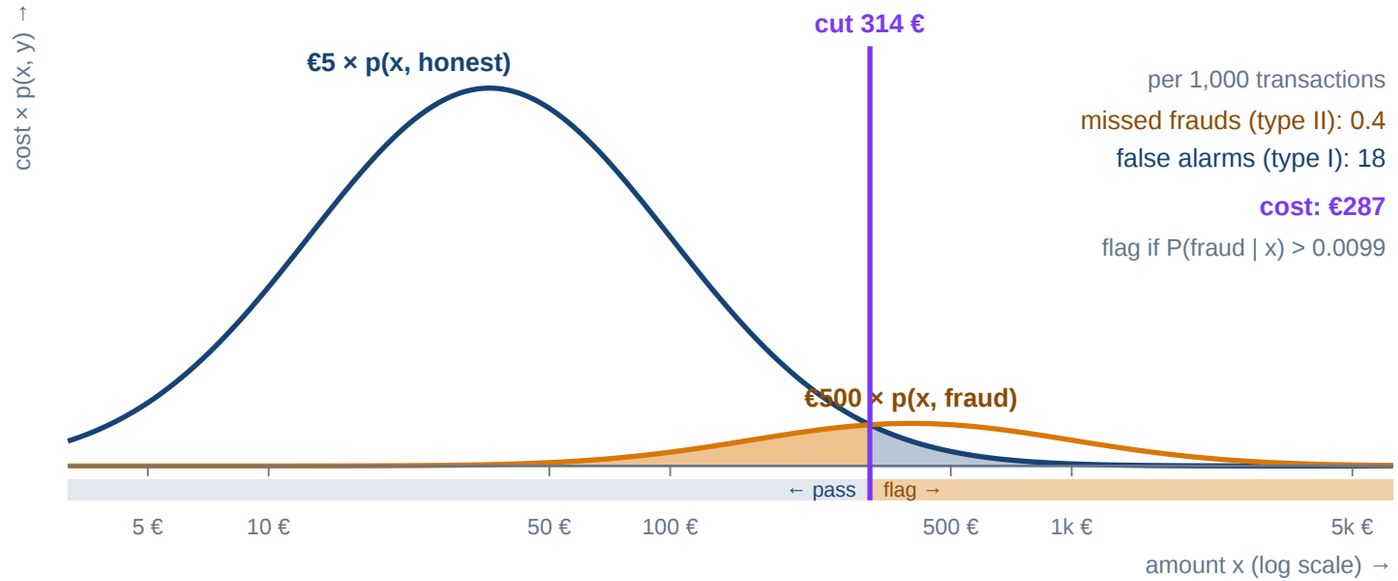
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$$\underbrace{(1-p)\,c_{FP}}_{\text{check}} < \underbrace{p\,c_{FN}}_{\text{let through}} \quad \Longleftrightarrow \quad p > p^* = \frac{c_{FP}}{c_{FP}+c_{FN}} = \frac{5}{505} \approx 1\%$$

> Lecture 01's loan threshold, for fraud. The three-flag transaction (17%) is not more likely fraud than not — and well worth a check.

Costs and Prior in One Threshold

Fraud 0.1%, a miss costs 100× — the cut where it was



$$\text{check iff} \quad \mathrm{LR}(x) > \frac{c_{FP}}{c_{FN}} \cdot \frac{P(\text{honest})}{P(\text{fraud})}$$

Predict before the last click: fraud becomes 100 times rarer. Where does the cut go?

Mini-Exercise

A transaction has likelihood ratio **20**; prior odds $1 : 999$.

Is it fraud under the Bayes decision rule? Is it worth a €5 check against a €500 loss?

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No, and yes. $20 < 999$: $P(\text{fraud} \mid x) = 20/1{,}019 \approx 2\%$. The check pays above 1%: on the ratio, above $999 \cdot 5/500 \approx 10$.

Takeaway

Deciding the **most probable class** gives the fewest errors — the **Bayes decision rule**; what remains is the overlap of the classes. **Unequal costs** move the threshold on the posterior to $c_{FP}/(c_{FP} + c_{FN})$. For a generative model costs and **prior** meet in one threshold on the likelihood ratio: both change without refitting $p(x \mid y)$.